Summary Report

Volleyball-Group-Activity-Recognition

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Abstract—A condensed summary of the Volleyball Group Activity Recognition project: the data pipeline, the generic multi-baseline loader, and the four completed baselines — B1, a single-frame fine-tuned ResNet-50 (62.6% test accuracy / 0.630 macro F1); B3, a person-then-group crop classifier (60.7% / 0.589); B4, a temporal classifier that runs an LSTM over B1's frozen frame features (66.1% / 0.673, best macro F1); and B5, a per-player temporal model that pools LSTM summaries of each player's crop sequence (66.3% / 0.619, best accuracy). The dataset (Ibrahim et al., CVPR 2016) contains 55 videos / 4,830 clips with 8 scene-level group activities and 9 player-level actions.

Keywords— *Volleyball, Group Activity Recognition, CNN, RNN, LSTM, ResNet50, Classification, Multiclass Classification, Transfer Learning*

1. Problem & Dataset

Volleyball Group Activity Recognition is a hierarchical classification problem: predict the scene-level group activity (8 classes: `l/r`-pass, -spike, -set, -winpoint) and the per-player action (9 classes: blocking, digging, falling, jumping, moving, setting, spiking, standing, waiting). The dataset comes from "A Hierarchical Deep Temporal Model for Group Activity Recognition" (Ibrahim et al., CVPR 2016; official repo, Kaggle mirror).

Key facts: **55 videos, 4,830 clips** (train/val/test = 2,152/1,341/1,337 clips over disjoint video splits of 24/15/16). Each clip has **41 frames**; the clip folder is named after its annotated *middle frame*. Both label levels are heavily skewed: `standing` is ≈68% of person actions, and `l/r_winpoint` are ≈2.5× rarer than the other group classes. The group classes come in **left/right mirror pairs**, so horizontal geometry is part of the label — the transform pipelines therefore use no horizontal flips and warp inputs directly to 224 × 224 so the full court (or the full player) always stays in view.

2. Data Pipeline

Three raw annotation sources (action detections, person detections, player tracking) are unified by a two-stage parser (`src/json_parser.py`) into one master JSON keyed `video_id/clip_id`, then enriched with the group-activity label per clip (looked up per video against each clip's own middle frame) and cached as a pickle (∼247 MB) for instant metadata loading. Frames are served either from a memory-mapped LMDB (local) or straight from disk (Kaggle); both backends expose the identical `VolleyballDataset`, which covers every baseline through flags: full frames or per-player crops, 1 or 9 frames per clip. A custom collate pads the variable player count per clip and returns a validity mask so padded slots never reach the loss or the pooling.

3. Baseline Roadmap

Table 1 lists the planned progression of models, each adding temporal and/or player-level structure on top of the previous one.

Table 1. Baseline models. B1, B3, B4, and B5 are complete; B6–B8 pending.

Model	Temporal	Player	Scene
B1	—	—	Image clf.
B3	—	Crop clf.	Concat-pool + MLP
B4	LSTM	—	LSTM
B5	LSTM/player	Concat-pool	MLP
B6	LSTM/image	Max-pool	LSTM
B7	$\text{LSTM}_1 + \text{LSTM}_2$	Max-pool	LSTM_2
B8	$\text{LSTM}_1 + \text{LSTM}_2$	Team-split	LSTM_2

4. Baseline 1: Single-Frame Classifier

Fine-tunes a pretrained ResNet-50 on the clip's middle frame for the 8 group classes. Two-stage schedule: linear probe (head only, 10 epochs, lr 10^{-3}), then partial fine-tune of `layer3/4` + head (up to 100 epochs, backbone lr 2.5×10^{-4}, head 3×, cosine annealing, batch 16, frozen BatchNorm held in eval).

Table 2. Baseline 1 test-set metrics.

Metric	Value
Accuracy	62.60%
Macro F1	0.630
Test Loss	1.42

Fig. 1 shows the per-class confusion on the test split.

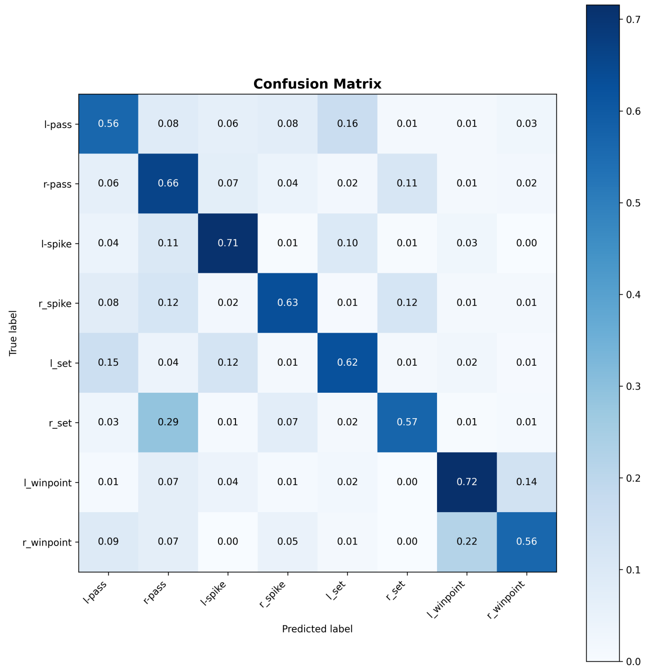

Figure 1. Baseline 1 test confusion matrix.

5. Baseline 3: Person-Then-Group

Stage A fine-tunes ResNet-50 on individual player crops for the 9 person actions with inverse-frequency class-weighted CE (without it the loss collapses onto `standing`). **Stage B** freezes that backbone, extracts a $(P, 2048)$ feature matrix per clip, concatenates masked max- and mean-pooling across players (2×2048), and trains an MLP ($4096 \to 512 \to 8$, Dropout 0.4) with class-weighted CE for the group classes. SGD (momentum 0.9, Nesterov), 50 epochs and patience 10 per stage, batch 16 clips, `nn.DataParallel` on multi-GPU hosts. Stage A reaches 70.4% validation accuracy (macro F1 0.512) on the person actions; Stage B 60.0% (0.584) on the group activities.

Fig. 2 shows the dominant error mode: left/right mirror confusion within each activity pair, worst on the rare `r_winpoint`. The main open issue is **overfitting** (train ≈95% vs. val 58–70%), to be addressed with stronger augmentation, higher dropout, and earlier stopping in a follow-up run.

Table 3. Baseline 3 test-set metrics.

Metric	Value
Accuracy	60.73%
Macro F1	0.589
Test Loss	1.09

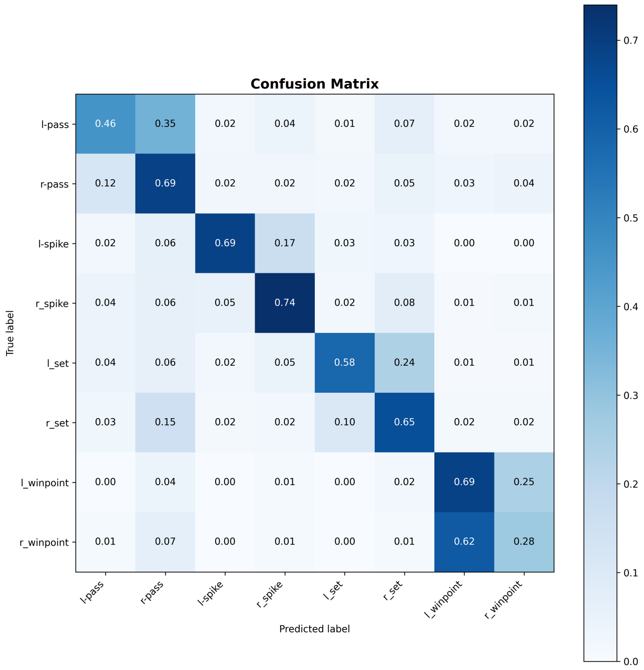

Figure 2. Baseline 3 test confusion matrix.

6. Baseline 4: Temporal Image Classifier

Each clip's 9-frame window is pushed through a **frozen** feature extractor loaded from Baseline 1's fine-tuned backbone, producing a (9, 2048) feature sequence per clip. A single-layer LSTM (hidden 512) consumes the sequence and its final hidden state is classified by a small MLP head (512 → 256 → 8, Dropout 0.3). Only the LSTM + head train (≈5.4M parameters; SGD lr 10^{-3}, cosine annealing, class-weighted CE, batch 16 clips).

Table 4. Baseline 4 test-set metrics.

Metric	Value
Accuracy	**66.12%**
Macro F1	**0.673**
Test Loss	1.05

B4 is the strongest baseline so far: temporal context over B1's own features is worth **+3.5 accuracy points / +0.043 macro F1** versus B1's single-frame result — the first direct evidence in this project that motion carries group-activity signal. Convergence is very fast (best validation F1 at epoch 4; training accuracy ≈99% by epoch 14, where early stopping fired), repeating the overfitting pattern of B1/B3. Fig. 3 shows the test confusion matrix.

7. Baseline 5: Temporal Person-Level Model

A two-stage temporal extension of B3. **Stage A** feeds each player's 9-crop sequence through B3's **frozen** Stage-A person-action backbone and trains a shared LSTM (hidden 3072, 1 layer) + head to classify the 9 person actions from the sequence's final hidden state. **Stage B** freezes the Stage-A model entirely, computes one LSTM summary

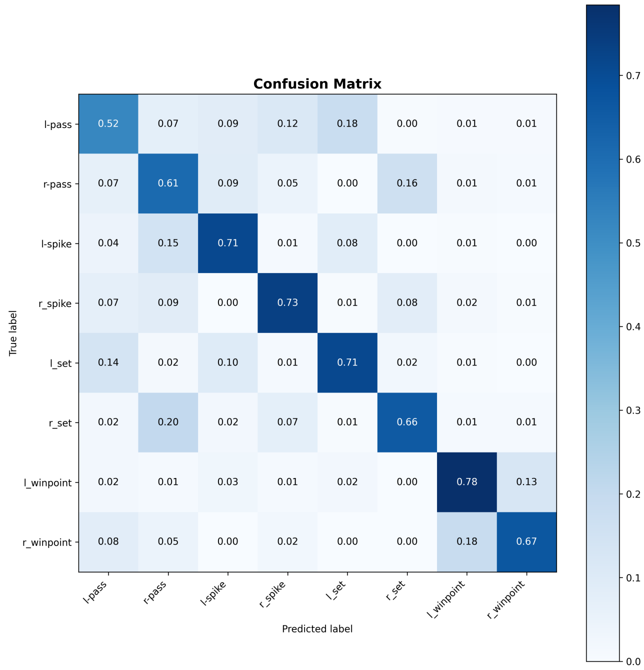

Figure 3. Baseline 4 test confusion matrix.

per player per clip, pools the summaries across players with masked concat (max || mean, classifier input 2 × 3072), and trains an MLP head (6144 → 3072 → 1536 → 8, LayerNorm + Dropout 0.2) on the 8 group activities. Training uses SGD (lr 10^{-3}, cosine annealing in Stage B), class-weighted CE in both stages, and gradient accumulation (effective batch 8 = micro 4 × 2) since each clip expands to 9 × ∼12 crop images.

Table 5. Baseline 5 test-set metrics.

Metric	Value
Accuracy	**66.34%**
Macro F1	**0.619**
Test Loss	**0.97**

B5 achieves the **best test accuracy and test loss of all baselines** (66.3% / 0.97 vs. B4's 66.1% / 1.05) and is markedly stronger on the core activities — spike F1 reaches 0.78–0.80 and set 0.71 on both court sides, with cross-activity confusion nearly eliminated — direct evidence that per-player temporal modeling adds signal beyond B4's scene-level features. Its macro F1 (0.619) nevertheless trails B4 (0.673) because of a single failure mode: **the `l/r_winpoint` confusion is the worst left/right confusion of any baseline in this project**. As Fig. 4 shows, `r_winpoint` collapses to 0.08 recall (F1 0.13), with 83% of true `r_winpoint` clips predicted as `l_winpoint`. The cause is structural: orderless max/mean pooling over all ∼12 players erases which side of the net each player occupies; winpoint clips show both teams in near-identical celebration poses, so side identity is the only discriminative signal — and the pooling destroys it. B8's team-split pooling (pool each team's six players separately, then concatenate) directly targets this failure.

8. Next Steps

(i) Reduce overfitting across all baselines — stronger augmentation, higher dropout, and earlier stopping remain the levers (B5's Stage B still reaches ∼90% train accuracy against ∼67% validation); (ii) proceed to the remaining temporal baselines B6–B8 — above all B8, whose team-split pooling is the direct fix for B5's left/right winpoint

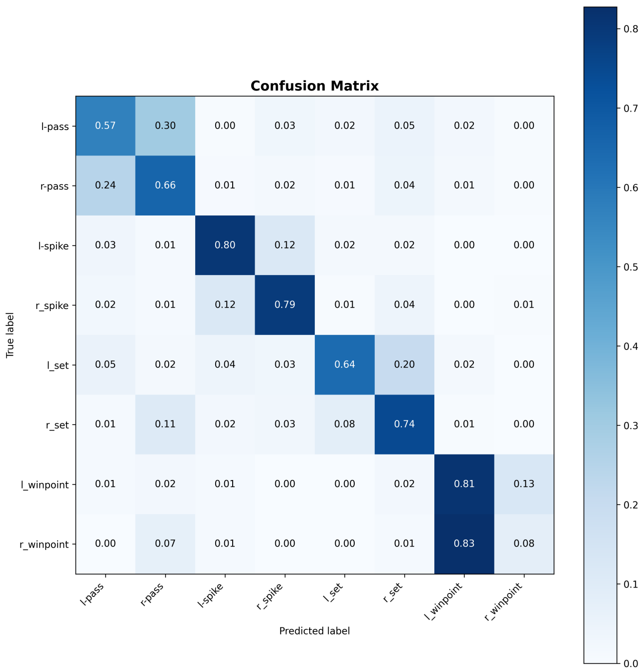

Figure 4. Baseline 5 test confusion matrix. Note the `r_winpoint` row: 83% of its clips are predicted as `l_winpoint` — the side-blind pooling failure discussed in the text.

collapse; the tracking boxes' x-coordinates already provide the team split.